Concern to Measure

Convert stakeholder language into structured concerns, scenarios, and measurable requirement instances. Follows ISO/IEC/IEEE 42010:2011 and TOGAF / BCS guidance. Allow 10-20 minutes per statement.

A	B	C	D	E	F
Raw statement (verbatim)	Classify concern type	Reframe as a concern	Scenario (env / stim / resp)	Measure (metric + evidence)	Requirement instance (SMART)

Figure: Raw statement → Classify → Reframe → Scenario → Measure → Requirement instance

1 · Raw Statement

Who / role: ______________________________ **Date:** ______________________ **System / service:** ______________________________

Raw statement (verbatim):

Context -- when / where / which user journey or business scenario:

Include environment assumptions: peak / normal / degraded · locations · user groups.

Impact -- what is at risk (harm / cost / delay / compliance / operations):

2 · Classify Concern Type [tick all that apply]

Functional attributes -- what the system must do

- ■ Workflow / process step (create, approve, fulfil, reconcile)
- ■ Data capture / validation / calculations
- ■ Reporting / analytics output
- ■ Integration / API / interface behaviour
- ■ Roles / permissions / approvals (as a functional need)
- ■ Notifications / messaging
- ■ Search / retrieval
- ■ Audit trail as a feature (who did what, when)
- ■ Other: ____________________________________

Quality attributes / NFR types -- how well it must do it *(ISO/IEC 25010 primary labels → BCS NFR terms)*

- ■ Performance efficiency (ISO 25010) → Performance (BCS: throughput / response time)
- ■ Reliability (ISO 25010) → Availability / reliability / recoverability (BCS)
- ■ Security (ISO 25010) → Security / privacy / information assurance (BCS)
- ■ Usability (ISO 25010) → Usability / customer experience (accessibility often sits here)
- ■ Compatibility (ISO 25010) → Interoperability / integratability (BCS)
- ■ Maintainability (ISO 25010) → Maintainability / modifiability / evolvability (BCS)
- ■ Portability (ISO 25010) → Portability (BCS)

Tip: keep type vs instance clear -- e.g. 'throughput' as type vs '100 tps' as instance (Avancier).

Planning & governance attributes -- constraints, risks, controls *(ISO 42010 §4.2)*

- ■ Cost / affordability
- ■ Schedule / deadline / time-to-market
- ■ Feasibility / known limitations
- ■ Regulatory / compliance constraint
- ■ Risk / assumptions / issues / dependencies (RAID prompts)
- ■ Standards / policy / principles constraint
- ■ Decision authority required (who must approve / sign off)
- ■ Supplier / contract / governance control requirement
- ■ Other: ____________________________________

■ Possible conflict (forces)? ■ Yes ■ No

If yes, name competing concerns and treat them as forces rather than personal disagreement.

3 · Reframe as a Concern [neutral, actionable]

Template: "Our concern is [topic] so that [stakeholder outcome], under [context]." -- ISO 42010 defines concern as a stakeholder interest; TOGAF emphasises concerns as roots of requirements decomposition.

Reframed concern:

4 · Scenario [make it refutable -- SEI QAW / ATAM structure]

Environment (assumptions):	*Load / degraded mode / location / user group*
Stimulus (event / load / failure / threat / change):	*What triggers the scenario?*
Scope (service / transaction / user group / component):	*Boundary of concern*
Response (observable behaviour):	*What the system does in response*
Response measure (how success is measured):	*Quantitative target -- feeds directly into the Requirement instance*

5 · Measure [requirement instance + evidence]

BCS: NFRs are usually quantitatively measurable; link each SMART "shall" statement to an acceptance test, monitoring threshold, or audit criterion.

Requirement instance (SMART "shall" statement):

Metric(s) & target(s):	**Measurement point (where measured):**
Time window / conditions:	**Owner:** _________________ **Due / gate:** _____________

Evidence method: ■ Test ■ Monitoring ■ Audit ■ Drill ■ Other: __________

References: ISO/IEC/IEEE 42010:2011 · ISO/IEC 25010:2011 · TOGAF® (The Open Group) · BCS Non-Functional Requirements guide · SEI Quality Attribute Workshop (QAW) / ATAM · Avancier Limited NFR guidance